UNIT I

Security Concepts & Cryptography

Cyber Security — Study Material for B.Tech Engineering Students

Subject: Cyber Security / Network Security

Syllabus Topics Covered
SECTION A — Security Concepts:
1. Introduction to Security 2. The Need for Security
3. Security Approaches 4. Principles of Security (CIA Triad + Extensions)
5. Types of Security Attacks 6. Security Services (X.800)
7. Security Mechanisms 8. A Model for Network Security
SECTION B — Cryptography Concepts and Techniques:
9. Introduction to Cryptography 10. Plain Text and Cipher Text
11. Substitution Techniques (Caesar, Monoalphabetic, Polyalphabetic, Playfair, One-Time Pad)
12. Transposition Techniques (Rail Fence, Columnar) 13. Encryption and Decryption
14. Symmetric and Asymmetric Key Cryptography 15. Steganography
16. Key Range and Key Size 17. Possible Types of Attacks on Cryptosystems

SECTION A
SECURITY CONCEPTS

1. Introduction to Security

Information security refers to the protection of information and information systems from unauthorised access, use, disclosure, disruption, modification, or destruction. In a world where every individual, organisation, and government depends on digital information, security is no longer optional — it is a fundamental engineering concern.

Security is a broad and multi-layered discipline. It encompasses physical security (protecting hardware), operational security (policies and procedures), network security (protecting data in transit), and application security (protecting software). Each layer must be addressed because a chain is only as strong as its weakest link.

2. The Need for Security

Need for Network Security:

1. Protects confidential information.
2. Maintains data integrity.
3. Provides user authentication.
4. Ensures availability of network resources.
5. Prevents unauthorized access.
6. Protects against cyber attacks.
7. Secures online transactions.
8. Protects organizational data and systems.
9. Enables secure communication.
10. Builds trust in online services.

Security is implemented through complementary layers of OSI. No single approach is sufficient; effective security combines all of them in a strategy .

Approach	Description	Example Controls
Prevention	Stop attacks before they occur	Firewalls, encryption, access control, patch management
Detection	Identify attacks as they occur or shortly after	Intrusion Detection Systems (IDS), log monitoring, SIEM tools

Response	React to and contain attacks in progress	Incident response plans, network isolation, emergency patching
Recovery	Restore normal operations after an attack	Backups, disaster recovery plans, business continuity planning

4. Principles of Security

4.1 The CIA Triad — Core Principles

The CIA Triad is the foundational model of information security. Every security goal, policy, and control can be traced back to one or more of these three principles.

Principle	Definition	Goal	Attack that Violates It	Countermeasure
Confidentiality	Ensuring information is accessible only to authorised parties	Prevent unauthorised disclosure	Eavesdropping, data theft, social engineering	Encryption, access control, data classification
Integrity	Ensuring information is accurate, complete, and unaltered	Prevent unauthorised modification	Man-in-the-middle attacks, data tampering	Hash functions, digital signatures, checksums
Availability	Ensuring systems and data are accessible to authorised users when needed	Prevent disruption of service	DoS/DDoS attacks, ransomware	Redundancy, backups, DDoS mitigation

4.2 Extended Principles (Beyond CIA)

Principle	Definition	Importance
Authentication	Verifying the identity of a user, system, or process	Ensures only legitimate users gain access; prevents impersonation
Authorisation	Determining what an authenticated entity is allowed to do	Enforces least privilege; limits damage from compromised accounts

Non-Repudiation	Ensuring a party cannot deny having performed an action	Critical for e-commerce, digital contracts, and audit trails
Accountability	Tracking user actions so they can be traced to an individual	Enables forensic investigation and deters insider attacks
Privacy	Protecting personal information from unauthorised collection or use	Legal requirement under GDPR, India's DPDP Act
Reliability	System consistently performs its intended function without failure	Ensures security controls themselves don't fail or can be bypassed

5. Types of Security Attacks

Security attacks are classified by the ITU-T X.800 standard into two broad categories: Passive Attacks and Active Attacks. Understanding this classification is essential for designing appropriate countermeasures.

5.1 Passive Attacks

Passive attacks involve monitoring or eavesdropping on transmissions without modifying the data. The attacker learns information from transmissions but does not alter them. They are difficult to detect because they leave no trace.

Passive Attack Type	Description	Goal	Detection Difficulty
Eavesdropping / Sniffing	Capturing network traffic using packet sniffers (e.g., Wireshark)	Steal credentials, sensitive data, session tokens	Very Hard — no system impact
Traffic Analysis	Analysing patterns of communication (who, when, how much) even if content is encrypted	Infer sensitive relationships and activities from metadata	Hard — requires statistical analysis to detect
Release of Message Contents	Reading the actual content of an unencrypted communication	Steal confidential information	Hard if no encryption used

Countermeasures for Passive Attacks: Encryption (makes captured data unreadable); VPNs (hides traffic content and metadata); TLS/SSL (encrypts web traffic); traffic padding (obscures traffic patterns)

5.2 Active Attacks

Active attacks involve modification of the data stream or creation of a false stream. They are easier to detect but harder to prevent because they require active participation by the attacker.

Active Attack Type	Description	CIA Principle Violated	Example
Masquerade (Spoofing)	An entity pretends to be a different (usually trusted) entity	Authentication, Confidentiality	IP spoofing, email phishing, ARP poisoning
Replay Attack	Capturing a valid transmission and replaying it later to gain unauthorised access	Authentication, Integrity	Replaying an authentication token to bypass login
Message Modification	Altering, delaying, reordering, or deleting parts of legitimate messages	Integrity	Man-in-the-middle changing a bank transfer amount
Denial of Service (DoS)	Overwhelming a system with requests to make it unavailable to legitimate users	Availability	Flooding a web server with millions of fake requests
Distributed DoS (DDoS)	DoS attack launched from many systems (botnet) simultaneously	Availability	2016 Mirai botnet DDoS took down Twitter, Netflix, GitHub
Malware Injection	Inserting malicious software into a system or network	All three — CIA	SQL injection, cross-site scripting (XSS), ransomware
Brute Force Attack	Trying all possible passwords or keys until the correct one is found	Confidentiality, Authentication	Automated password cracking tools like Hydra, Hashcat

Man-in-the-Middle (MITM)	Attacker secretly intercepts and relays communication between two parties	Confidentiality, Integrity	Attacker intercepts HTTPS session on public Wi-Fi

6. Security Services (X.800 Standard)

The ITU-T X.800 standard defines security services as services that enhance the security of data processing systems and information transfers of an organisation. These services are designed to counter security attacks and implement security policies.

Security Service	Definition	Mechanism Used	Counters Which Attack
Authentication	Verifying the identity of communicating entities (peer entity authentication) or the source of data (data origin authentication)	Digital signatures, passwords, certificates, biometrics	Masquerade, replay attacks
Access Control	Preventing unauthorised use of resources — controlling who can use what, under what conditions	ACLs, RBAC, capability lists, firewalls	Unauthorised access, privilege escalation
Data Confidentiality	Protecting data from passive attacks — ensuring only authorised parties can read the data	Encryption (AES, RSA), VPNs, TLS	Eavesdropping, traffic analysis
Data Integrity	Ensuring data has not been altered, destroyed, or modified in an unauthorised manner	Hash functions (SHA-256), MACs, digital signatures	Message modification, replay attacks
Non-Repudiation	Preventing either party from denying that they sent or received a message	Digital signatures, timestamps, audit logs	Denial of actions, fraud
Availability Service	Ensuring systems and data are accessible and usable on demand by authorised users	Redundancy, failover, DDoS protection, backups	DoS/DDoS attacks, ransomware

7. Security Mechanisms

Security mechanisms are tools, techniques, and procedures deployed to implement security services. The X.800 standard classifies them as specific (designed to implement a particular service) and pervasive (general mechanisms not specific to one service).

7.1 Specific Security Mechanisms

Mechanism	Description	Security Service It Supports
Encipherment (Encryption)	Transforming data using an algorithm and key to make it unintelligible to unauthorised parties	Confidentiality, limited integrity
Digital Signature	A cryptographic technique where the sender signs data with their private key; receiver verifies with public key	Authentication, integrity, non-repudiation
Access Control	Mechanisms enforcing authorisation policies — who can access what resources	Access control, authentication
Data Integrity Mechanisms	Hash functions and MACs that detect unauthorised modification	Data integrity
Authentication Exchange	Protocols where entities prove identity through challenge-response or certificates	Authentication
Traffic Padding	Inserting dummy data into traffic to confuse traffic analysis attempts	Confidentiality (against traffic analysis)
Routing Control	Selecting secure, trusted routes for data transmission; avoiding compromised paths	Confidentiality, availability
Notarisation	Using a trusted third party to assure properties of data exchange (time, origin, destination)	Non-repudiation, integrity

8. A Model for Network Security

The generic model for network security, developed by William Stallings, provides a framework for understanding how security is applied to communications between two parties. It identifies the key elements of any secure communication system.

8.1 The Generic Network Security Model

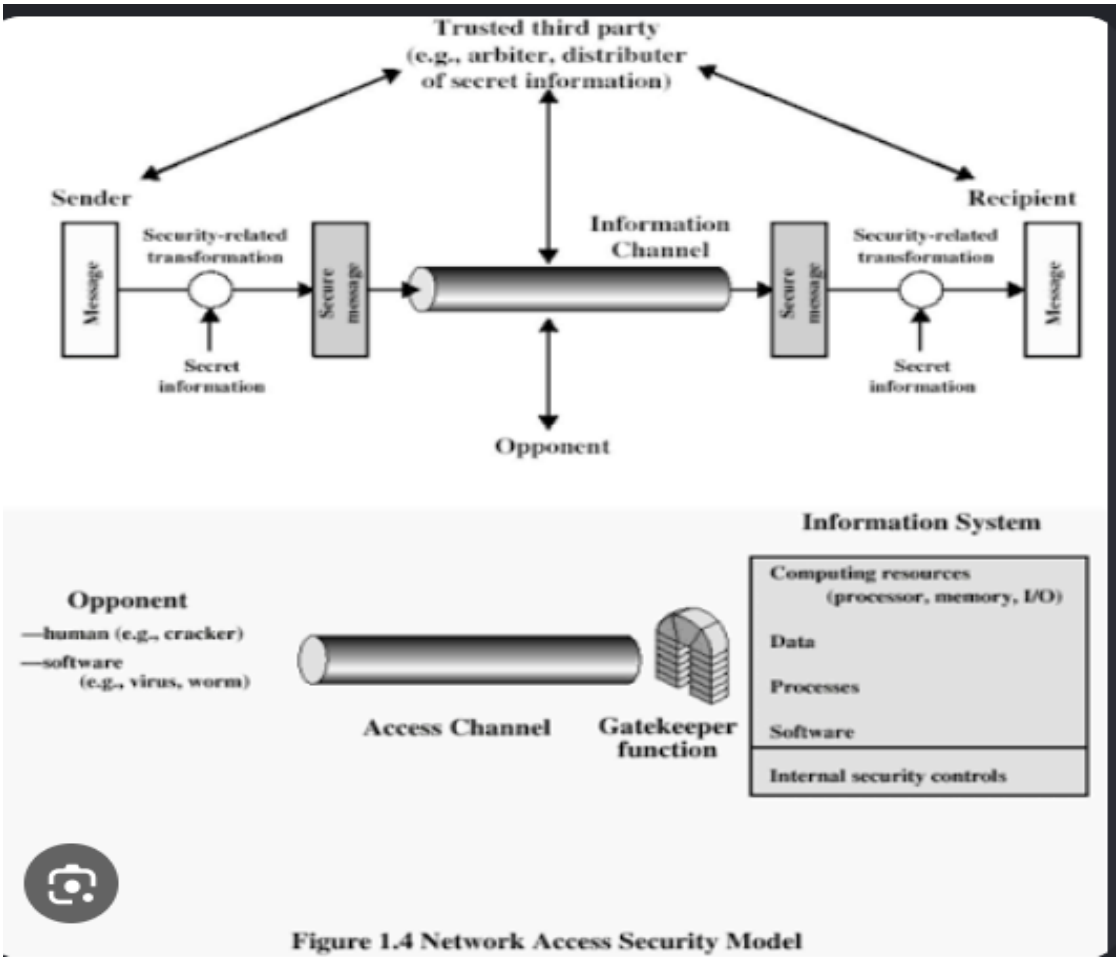


Figure 1.4 Network Access Security Model

8.2 How the Model Works

1. The sender has a message to send to the receiver through an insecure channel (e.g., the internet)
2. A security transformation (e.g., encryption) is applied using a secret key to produce a secure message
3. The secure message is transmitted through the channel where an opponent may attempt to intercept it
4. At the receiver, the inverse security transformation (e.g., decryption) is applied using the key
5. The original message is recovered if and only if the correct key is available
6. The Trusted Third Party (e.g., a Certificate Authority) may be involved in key distribution or identity verification

8.3 Design Challenges Revealed by the Model

- Algorithm Design: The security transformation algorithm must be mathematically strong
- Key Generation: Keys must be sufficiently large and randomly generated
- Key Distribution: Getting the secret key to sender and receiver securely is the hardest problem
- Key Management: Keys must be stored, rotated, and revoked securely throughout their lifecycle

SECTION B
CRYPTOGRAPHY CONCEPTS AND TECHNIQUES

9. Introduction to Cryptography

Cryptography (from Greek: kryptos = hidden, graphia = writing) is the science and art of securing information by transforming it into an unreadable format that can only be decoded by authorised parties. It is the cornerstone technology behind all modern information security.

Term	Definition
Cryptography	The science of designing algorithms and protocols to secure information
Cryptanalysis	The science of analysing and breaking cryptographic systems without the key
Cryptology	The umbrella field encompassing both cryptography and cryptanalysis
Cipher	An algorithm for performing encryption and decryption
Key	A piece of information (parameter) that controls the encryption/decryption process
Plaintext (P)	The original, readable message before encryption
Ciphertext (C)	The transformed, unreadable message after encryption
Encryption (E)	The process of converting plaintext to ciphertext: C = E(K, P)
Decryption (D)	The process of converting ciphertext back to plaintext: P = D(K, C)
Cryptosystem	A complete system consisting of algorithm + key space + plaintext space + ciphertext space

10. Plain Text and Cipher Text

The central transformation in cryptography is converting plaintext into ciphertext (encryption) and back (decryption).

Property	Plaintext	Ciphertext
Definition	Original, readable message understood by anyone	Encrypted, unreadable message — appears as random data
Representation	Human-readable text, binary data, or structured data	Scrambled characters, binary output, or encoded data

Example	'HELLO WORLD'	'KHOOR ZRUOG' (after Caesar Cipher with shift 3)

Mathematical Notation

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Encryption:  C = E(K, P)   — Ciphertext = Encrypt(Key, Plaintext)
Decryption:  P = D(K, C)   — Plaintext  = Decrypt(Key, Ciphertext)
For symmetric: Same key K used for E and D
For asymmetric: Public key Kpub for E; Private key Kpriv for D
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Types of Ciphers by Operation
Stream Cipher: Encrypts plaintext one bit/byte at a time using a keystream (e.g., RC4)
Block Cipher: Encrypts fixed-size blocks of plaintext at a time (e.g., AES operates on 128-bit blocks)
Substitution Cipher: Replaces each character with another character or symbol
Transposition Cipher: Rearranges (permutes) the positions of characters without changing them

11. Substitution Techniques

Substitution ciphers replace each character (or group of characters) of the plaintext with another character (or group). The positions of characters do not change — only their identity changes.

11.1 Caesar Cipher

The Caesar Cipher, attributed to Julius Caesar (~50 BCE), is the simplest and most widely known substitution cipher. Each letter is shifted by a fixed number of positions down the alphabet.

Encryption Formula: C = (P + K) mod 26 where P = plaintext letter position (A=0, B=1, ..., Z=25), K = shift value

Decryption Formula: P = (C - K + 26) mod 26

Caesar Cipher — Worked Example
Key (Shift): K = 3
Plaintext: H E L L O
Positions: 7 4 11 11 14
After +3: 10 7 14 14 17

Ciphertext: K H O O R
Full Alphabet Mapping (Shift 3):
Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C
Decryption of 'KHOOR' → Subtract 3 from each → 'HELLO'

Strengths and Weaknesses of Caesar Cipher

Aspect	Details
Key Space	Only 25 possible keys (shifts 1-25) — can be broken by brute force in seconds
Attack Method	Brute force: try all 25 shifts; frequency analysis: letter 'E' is most common in English
Security Level	Extremely weak — used only for educational purposes today
Historical Use	Used by Julius Caesar for military communications; once considered secure

11.2 Monoalphabetic Substitution Cipher

A Monoalphabetic cipher uses a single fixed substitution — each letter of the plaintext always maps to the same ciphertext letter, but the mapping is arbitrary (not a simple shift). This is a permutation of the alphabet.

Monoalphabetic Cipher — Example
Key (Substitution Alphabet):
Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Cipher: Q W E R T Y U I O P A S D F G H J K L Z X C V B N M
Plaintext: H E L L O
Ciphertext: I T S S G
Key Space: 26! ≈ 4 × 10^26 possible keys (astronomically large)

However: Vulnerable to frequency analysis because each letter always maps to the same cipher letter

Frequency Analysis Attack: In English, 'E' appears ~12.7% of the time. The most frequent ciphertext letter is likely 'E'. Letter bigrams (TH, HE, IN) and trigrams (THE, AND, ING) further narrow down the mapping.

11.3 Polyalphabetic Substitution Cipher — Vigenère Cipher

The Vigenère Cipher, invented by Giovan Bellaso (1553) and misattributed to Vigenère, uses multiple Caesar ciphers in sequence. A keyword determines which shift to apply to each letter, making frequency analysis much harder.

Encryption: Ci = (Pi + Ki) mod 26 where Ki is the ith letter of the repeating keyword
Decryption: Pi = (Ci - Ki + 26) mod 26

Vigenère Cipher — Worked Example
Keyword: KEY (K=10, E=4, Y=24 — repeats as needed)
Plaintext: A T T A C K A T D A W N
Keyword: K E Y K E Y K E Y K E Y
Key values: 10 4 24 10 4 24 10 4 24 10 4 24
P values: 0 19 19 0 2 10 0 19 3 0 22 13
P+K mod 26: 10 23 17 10 6 8 10 23 1 10 0 11
Ciphertext: K X R K G I K X B K A L

Feature	Caesar Cipher	Vigenère Cipher
Number of Alphabets	1 (single shift)	Multiple (length of keyword)
Key	Single number (1-25)	A word or phrase
Frequency Analysis	Trivially broken	Defeated for short keywords; broken by Kasiski test for long texts
Security	Very weak	Moderate (historically called 'le chiffre indéchiffrable')

11.4 Playfair Cipher

The Playfair Cipher, invented by Charles Wheatstone in 1854 and promoted by Lord Playfair, was the first practical digraph (two-letter) substitution cipher. It encrypts pairs of letters using a 5×5 keyword matrix.

Playfair Key Matrix Construction

7. Choose a keyword (e.g., MONARCHY) and write it out, removing duplicate letters
8. Fill the remaining cells of a 5×5 grid with the unused alphabet letters (I and J are combined into one cell)
9. Result is a 5×5 matrix with all 25 letters (I/J share one cell)

Playfair Matrix for Keyword 'MONARCHY'
M O N A R
C H Y B D
E F G I/J K
L P Q S T
U V W X Z

Playfair Encryption Rules

10. Split plaintext into digraphs (pairs). If both letters in a pair are the same, insert 'X' between them. Pad with 'X' if odd length.
11. Same Row: Each letter is replaced by the letter to its right (wrap around)
12. Same Column: Each letter is replaced by the letter below it (wrap around)
13. Rectangle Rule: Each letter is replaced by the letter in the same row but in the column of the other letter

Playfair Encryption Example — 'AR' using MONARCHY matrix
Plaintext pair: A R
A is at row 0, col 3. R is at row 0, col 4.
Same Row Rule: Replace each with the letter to its right (wrapping).
A (row 0, col 3) → col 4 = R
R (row 0, col 4) → col 0 (wrap) = M
Ciphertext: R M

Key Space: Playfair has 25! / 2 ≈ 1.5 × 10^25 possible keys — much stronger than Caesar or simple monoalphabetic

Weakness: Still vulnerable to frequency analysis on digraphs; 'TH' is the most common English digraph

11.5 One-Time Pad (Vernam Cipher)

The One-Time Pad (OTP), developed by Gilbert Vernam in 1917 and proven perfectly secure by Claude Shannon in 1949, is the ONLY cipher that is mathematically proven to be unconditionally secure — if used correctly.

How the One-Time Pad Works

14. Generate a truly random key that is AT LEAST as long as the plaintext
15. Encrypt: XOR each bit of plaintext with the corresponding bit of the key: C = P XOR K
16. Decrypt: XOR the ciphertext with the same key: P = C XOR K
17. The key must be used EXACTLY ONCE and then destroyed — hence 'one-time' pad

One-Time Pad — Worked Example (Letters)
Plaintext: H E L L O (positions: 7, 4, 11, 11, 14)
Key: X M C K L (positions: 23, 12, 2, 10, 11) — truly random
C = (P+K) mod 26: 30, 16, 13, 21, 25 → mod 26 → 4, 16, 13, 21, 25
Ciphertext: E Q N V Z
Security Proof: For any ciphertext, any plaintext is equally likely — zero information leaked

Condition for Perfect Secrecy	Consequence if Violated
Key must be truly random (not pseudo-random)	Predictable key allows attacker to reconstruct plaintext
Key must be as long as the plaintext	Shorter key reuse leads to statistical attacks
Key must never be reused	Two ciphertexts XORed together cancel the key, revealing plaintext relationship
Key must be kept completely secret	If key is known, OTP provides zero protection

Practical Limitation: The OTP requires securely sharing a key as large as all future messages — impractical at scale. Used for top-secret diplomatic communications (e.g., Moscow-Washington hotline).

12. Transposition Techniques

Transposition ciphers rearrange (permute) the characters of the plaintext according to a fixed rule or key. Unlike substitution ciphers, the characters themselves are not changed — only their positions change. The same set of letters appears in the ciphertext; only the order differs.

12.1 Rail Fence Cipher

The Rail Fence cipher is a simple transposition technique where the plaintext is written diagonally across a number of 'rails' (rows) in a zigzag pattern, then read off row by row to produce the ciphertext.

Rail Fence Cipher — Worked Example
Plaintext: ATTACK AT DAWN (remove spaces: ATTACKATDAWN)
Key: 3 rails (rows)
Write diagonally across 3 rails:
Rail 1: A . . . C . . . D . . .
Rail 2: . T . A . K . T . A . N
Rail 3: . . T . . . A . . . W .
Read each rail left to right:
Rail 1: A C D
Rail 2: T A K T A N
Rail 3: T A W
Ciphertext: ACD + TAKTAN + TAW = ACDTAKTANTAW
Decryption: Reconstruct the zigzag pattern with the same number of rails, fill in ciphertext, read diagonally

12.2 Columnar Transposition Cipher

The Columnar Transposition cipher writes the plaintext in rows of a fixed width (determined by the key length), then reads off the columns in an order specified by the key.

Columnar Transposition — Worked Example

Plaintext: MEET ME AFTER THE TOGA PARTY (remove spaces: MEETMEAFTERTHETOGAPARTY)
Key: 4 3 1 2 5 6 7 (key word MEGABYTE; alphabetical order of letters determines column order)
For simplicity, use key = HACK (H=2, A=1, C=3, K=4 → column order: 2,1,3,4)
Key Letters: H A C K
Col order: 2 1 3 4

Write plaintext row by row (4 columns):
Row 1: M E E T
Row 2: M E A F
Row 3: T E R T
Row 4: H E T O
Row 5: G A P A
Row 6: R T Y X (pad with X)

Read columns in order 1,2,3,4 (A,H,C,K):
Col 1 (A): E E E E A T
Col 2 (H): M M T H G R
Col 3 (C): E A R T P Y
Col 4 (K): T F T O A X
Ciphertext: EEEEAT MMTHGR EARTPY TFTOA X

Feature	Substitution Ciphers	Transposition Ciphers
What Changes	The identity of characters (A→K)	The position of characters (A stays A, moves place)
What Stays Same	The position of characters	The set of characters (same letters, different order)
Attack Method	Frequency analysis of individual letters/digraphs	Anagramming; frequency analysis of groups
Examples	Caesar, Vigenère, Playfair, OTP	Rail Fence, Columnar, Route Cipher

Combined Use	Product Cipher: applying both substitution AND transposition significantly increases security	DES and AES use multiple rounds of both techniques

13. Encryption and Decryption

Encryption is the process of transforming intelligible plaintext into unintelligible ciphertext using a cipher algorithm and a key. Decryption is the reverse process — recovering the plaintext from ciphertext using the same algorithm (and same or paired key).

Formal Definition

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Encryption:  C = Ek(P)  — Apply encryption function E with key k to plaintext P
Decryption:  P = Dk(C)  — Apply decryption function D with key k to ciphertext C
Correctness: Dk(Ek(P)) = P  — Decryption of encryption returns original plaintext
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Properties of a Good Encryption Scheme

- Completeness: Every plaintext bit should influence every ciphertext bit (diffusion)
- Avalanche Effect: A small change in plaintext or key causes a large change in ciphertext
- Confusion: The relationship between key and ciphertext should be complex (substitutions)
- Diffusion: The influence of each plaintext bit should spread across many ciphertext bits (transpositions)
- Non-linearity: The encryption function must not be expressible as a linear equation
- Key Sensitivity: Different keys should produce completely different ciphertexts for the same plaintext

14. Symmetric and Asymmetric Key Cryptography

14.1 Symmetric Key Cryptography

Symmetric key cryptography (also called secret key or conventional cryptography) uses the SAME key for both encryption and decryption. Both communicating parties must share the secret key in advance through a secure channel.

How Symmetric Encryption Works

Symmetric Key Encryption Model
1. Alice and Bob agree on a secret key K (through a secure channel)

2. Alice encrypts message M: C = Ek(M)
3. Alice sends ciphertext C over the insecure network
4. Bob decrypts: M = Dk(C) using the same key K
5. Eve intercepts C but cannot decrypt without K

Common Symmetric Algorithms

Algorithm	Key Size	Block Size	Year	Notes
DES	56 bits	64 bits	1977	Now broken — 56-bit key crackable in hours by brute force
AES (Rijndael)	128/192/256 bits	128 bits	2001	Current global standard; used in TLS, Wi-Fi, disk encryption
RC4	40-2048 bits	Stream	1987	Stream cipher; used in SSL/WEP but now deprecated (vulnerable)
Blowfish	32-448 bits	64 bits	1993	Fast, free; used in older systems and bcrypt password hashing

Aspect	Symmetric Key Cryptography
Keys	Single shared secret key for encryption and decryption
Speed	Fast — suitable for encrypting large volumes of data
Key Distribution Problem	Hard — securely sharing the key is the biggest challenge
Key Management	N parties need N(N-1)/2 keys for all pairs — scales poorly
Examples	AES, DES, 3DES, RC4, Blowfish, ChaCha20
Best Used For	Bulk data encryption: disk encryption, VPN tunnels, file encryption
Authentication	Cannot provide non-repudiation — shared key means either party could have created the message

14.2 Asymmetric Key Cryptography (Public Key Cryptography)

Asymmetric cryptography, invented by Whitfield Diffie and Martin Hellman in 1976 and independently by Ralph Merkle, uses a KEY PAIR: a public key (freely shared) and a private key (kept secret by the owner). What one key encrypts, only the other key can decrypt.

How Asymmetric Encryption Works

Asymmetric Key Encryption — Confidentiality Model
1. Bob generates a key pair: (Public Key = Kpub, Private Key = Kpriv)
2. Bob publishes Kpub widely — anyone can have it
3. Alice encrypts message M using Bob's public key: C = E(Kpub_Bob, M)
4. Alice sends C over the insecure network
5. Only Bob can decrypt: M = D(Kpriv_Bob, C) — because only Bob has Kpriv
6. Eve has C and Kpub_Bob but cannot compute Kpriv_Bob (mathematically infeasible)

Asymmetric Key — Digital Signature Model (Authentication + Non-repudiation)
1. Alice signs message M with her private key: S = Sign(Kpriv_Alice, H(M))
2. Alice sends (M, S) to Bob
3. Bob verifies: Verify(Kpub_Alice, M, S) — uses Alice's public key
4. If verification passes, M came from Alice and was not modified
5. Alice cannot deny sending M — non-repudiation achieved

Common Asymmetric Algorithms

Algorithm	Basis of Security	Key Size	Primary Use
RSA	Difficulty of factoring large prime products	2048-4096 bits	Encryption, digital signatures, key exchange
Diffie-Hellman (DH)	Discrete logarithm problem	2048+ bits	Key exchange protocol (not directly encryption)
DSA (Digital Signature Algorithm)	Discrete logarithm	1024-3072 bits	Digital signatures only (not encryption)
Elliptic Curve Cryptography (ECC)	Elliptic curve discrete logarithm	256-521 bits	Efficient encryption and signatures; used in TLS, Bitcoin
ElGamal	Discrete logarithm	2048+ bits	Encryption and digital signatures

Hybrid Cryptography — The Best of Both Worlds

Modern systems (TLS, PGP, SSH) use HYBRID cryptography: asymmetric cryptography to securely exchange a symmetric session key, then symmetric cryptography for the bulk data transfer.

Hybrid Encryption — TLS Handshake Model
Step 1: Client sends supported cipher suites to server
Step 2: Server sends its X.509 certificate (contains server's public key)
Step 3: Client verifies certificate via Certificate Authority (CA)
Step 4: Client generates a random session key; encrypts it with server's public key (asymmetric)
Step 5: Server decrypts with its private key to recover session key
Step 6: Both sides now share the session key; all further communication uses AES (symmetric)
Result: Asymmetric solves key distribution; symmetric provides fast bulk encryption

15. Steganography

Steganography (from Greek: steganos = covered, graphia = writing) is the practice of hiding secret information within ordinary, non-secret data or files so that the existence of the secret message is concealed. Unlike cryptography which hides the content of a message, steganography hides the fact that a message exists at all.

Aspect	Cryptography	Steganography
Goal	Hide the CONTENT of the message	Hide the EXISTENCE of the message
Appearance	Ciphertext is obviously encrypted (suspicious)	Carrier medium looks completely normal
Detection	Attacker knows message exists but cannot read it	Attacker does not know a message exists
Combination	Often combined: encrypt then hide for maximum security	Alone: if carrier detected, message may be found
Examples	AES, RSA encryption	Hidden text in image LSBs, watermarks, invisible ink

Types and Techniques of Steganography

Type	Medium	Technique	Tool/Example
Image Steganography	JPEG, PNG, BMP files	Least Significant Bit (LSB) insertion: replace LSBs of pixel values with message bits	OpenStego, SilentEye, StegHide
Audio Steganography	MP3, WAV files	Echo hiding, phase coding, spread spectrum — hide data in inaudible frequency ranges	MP3Stego, DeepSound
Video Steganography	AVI, MP4 files	Hide data in individual frames — spatial or temporal redundancy	OpenPuff
Text Steganography	Plain text documents	Extra whitespace, misspelled words, first letter of each line spells a message	Manual or specialized tools
Network Steganography	IP/TCP packets	Unused packet header fields carry hidden data	Covert channel tools
Document Steganography	PDF, DOCX files	Invisible white text, metadata, font variations	Metadata tools

16. Possible Types of Attacks on Cryptosystems

Cryptanalysis is the science of breaking cryptographic systems. Attacks on cryptosystems are classified by the information available to the attacker (Kerckhoffs' Principle: assume the attacker knows the algorithm; only the key must be secret).

16.1 Classification by Attacker's Knowledge

Attack Type	What Attacker Knows	What Attacker Wants	Difficulty	Example
Ciphertext-Only Attack (COA)	Ciphertext only (C)	Recover plaintext P or key K	Hardest for attacker; most realistic scenario	Frequency analysis on Caesar Cipher
Known-Plaintext Attack (KPA)	Some (P, C) pairs — same key used	Find key K or decrypt other ciphertexts	Moderate — attacker has partial information	WWII Enigma broken using known German phrases

Chosen-Plaintext Attack (CPA)	Can choose any plaintext and get its encryption	Recover key or learn algorithm details	Easier — attacker controls input	Differential cryptanalysis of DES
Chosen-Ciphertext Attack (CCA)	Can choose any ciphertext and get its decryption	Recover key or plaintext of target ciphertexts	Even easier — strong threat model	Padding oracle attack on early TLS

QUICK REVISION: Key Concepts at a Glance

Topic	Key Points to Remember
CIA Triad	Confidentiality (encryption), Integrity (hashing/signatures), Availability (uptime/backups)
Passive Attacks	Eavesdropping, traffic analysis — read only, no modification — hard to detect
Active Attacks	Masquerade, replay, modification, DoS, MITM — modify or disrupt — easier to detect
X.800 Security Services	Authentication, Access Control, Confidentiality, Integrity, Non-Repudiation, Availability
Security Mechanisms	Encipherment, Digital Signature, Access Control, Integrity, Traffic Padding, Notarisation
Network Security Model	Sender → Security Transform → Channel → Security Transform → Receiver; Trusted Third Party
Caesar Cipher	Shift by K; C=(P+K) mod 26; only 25 keys; broken by brute force
Monoalphabetic	Random letter mapping; 26! keys; broken by frequency analysis
Vigenère	Repeating keyword; polyalphabetic; broken by Kasiski test
Playfair	5×5 matrix; digraph substitution; I/J combined; 3 rules (row/col/rectangle)
One-Time Pad	XOR with random key; unconditionally secure; key must be random, same length, used once
Rail Fence	Write zigzag across N rails; read row by row
Columnar Transposition	Write in rows; read columns in key-specified order
Symmetric Crypto	Same key encrypt/decrypt; fast; key distribution problem; AES is standard
Asymmetric Crypto	Public/private key pair; slow; solves key distribution; RSA/ECC
Hybrid Crypto	Asymmetric for key exchange + Symmetric for bulk data (TLS model)
Steganography	Hide existence of message (not just content); LSB in images; differs from cryptography
Key Size	2^n possible keys for n-bit key; 128 bits minimum today; AES-256 is quantum-safe
Brute Force	Try all keys; only defense is large key size
Frequency Analysis	Exploit letter frequency in natural language; defeated by polyalphabetic/modern ciphers

Sample Examination Questions

Long Answer Questions (10-15 marks)

18. Explain all X.800 security services and their corresponding security mechanisms with examples.
19. Compare symmetric and asymmetric key cryptography comprehensively..
20. Explain the Playfair Cipher: key matrix construction, encryption rules, and a worked example. What are its strengths over simple monoalphabetic ciphers?
21. Describe the Vigenère Cipher with a complete worked example. .
22. Classify and explain all possible attacks on cryptosystems. For each attack type, describe the attacker's knowledge, the goal, and a real-world countermeasure.
23. Explain the Rail Fence and Columnar Transposition ciphers with complete worked examples. How do transposition ciphers differ from substitution ciphers?

Formula and Reference Sheet

Cryptography Formulas

Formula	Meaning
C = (P + K) mod 26	Caesar Cipher Encryption (P=plaintext position, K=shift, C=ciphertext position)
P = (C - K + 26) mod 26	Caesar Cipher Decryption
Ci = (Pi + Ki) mod 26	Vigenère Cipher Encryption (Ki = ith keyword letter position)
Pi = (Ci - Ki + 26) mod 26	Vigenère Cipher Decryption
C = P XOR K	One-Time Pad Encryption (XOR operation)
P = C XOR K	One-Time Pad Decryption
C = E(Kpub, P)	Asymmetric Encryption (encrypt with recipient's public key)
P = D(Kpriv, C)	Asymmetric Decryption (decrypt with recipient's private key)
Key Space = 2^n	Number of possible keys for an n-bit key
Keys needed for N users (Symmetric) = N(N-1)/2	Total symmetric keys for a group of N users

Keys needed for N users (Asymmetric) = 2N	Total asymmetric keys (1 pair each)

Standard ASCII Letter Positions (for cipher calculations)

Letter Positions: A=0, B=1, C=2, D=3, E=4, F=5, G=6, H=7, I=8, J=9, K=10, L=11, M=12,
N=13, O=14, P=15, Q=16, R=17, S=18, T=19, U=20, V=21, W=22, X=23, Y=24, Z=25
Tip: Count from A=0 (not A=1) — this matches the mod 26 arithmetic used in cipher formulas

End of Unit I Study Material — Cyber Security